INCH-POUND
MIL-DTL-24643/83
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 300 VOLTS, TOPSIDE, 75 OHMS, MULTI-SHIELDED PAIRS, TYPES LSE1SPE, LSE2SPE, LSE3SPE, LSE4SPE, LSE6SPE, LSE7SPE, LSE8SPE, LSE10SPE, LSE12SPE, LSE14SPE, LSE16SPE, LSE19SPE, LSE24SPE, LSE30SPE, LSE37SPE, LSE44SPE, AND LSE61SPE (INCLUDING VARIANTS LSE1SPEW, LSE2SPEW, LSE3SPEW, LSE4SPEW, LSE6SPEW, LSE7SPEW, LSE8SPEW, LSE10SPEW, LSE12SPEW, LSE14SPEW, LSE16SPEW, LSE19SPEW, LSE24SPEW, LSE30SPEW, LSE37SPEW, LSE44SPEW, AND LSE61SPEW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSE**SPE (non-watertight) and LSE**SPEW (watertight).

Construction (Non-Watertight and Watertight)

First	-	Copper conductor, tin-coated, 22 AWG and 20 AWG in accordance with ASTM B286; 18 AWG and 16 AWG in accordance with ASTM B8 (see table I or table II for size).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Wall thickness as necessary, in order for components in completed cable to meet the electrical requirements specified herein.
Fourth	-	Two conductors, one black and one white (or natural), twisted together with a nominal lay of 2.5 inches to form a pair.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 36 or 38 AWG tin-coated copper. Braid angle of 30 to 35 degrees. Minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material. Minimum average thickness of the transparent material shall be 0.003 inch. Standard identification codes shall be applied by method 2 on the inner tape.
Eighth	-	The specified number of shielded pairs (see table I or table II) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Ninth	-	An optional binder.
Tenth	-	Copper foil/polyester tape. Applied helically with 50 percent minimum overlap (metallic foil side facing out and in contact with the braid).

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Eleventh - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

Twelfth - Copper foil/polyester tape/copper foil. Applied helically with 50 percent minimum overlap. Tape shall provide electrical contact between the inner and outer braided shields.

Thirteenth - Braided shield of tin-coated copper. Braid shall be constructed such that the completed cable complies with the surface transfer impedance requirements specified herein. In addition, the minimum coverage of the braid shall be 90 percent and the minimum shield strand size shall be 34 AWG.

For watertight constructions only, the use of solid or semi-solid water-block materials (such as silicone, hot melts, or gels) is not allowed on the outer braid. Watertightness shall be by means that could include, but are not limited to, water-swell yarns incorporated within or over the braid or water-swell tape helically applied over the outer braid with 10 percent minimum overlap. The water-block material shall not stick or adhere to the outer braid strands.

Fourteenth - An optional binder (tape shall not be clear).

Fifteenth - Cross-linked polyolefin jacket. See table I or table II for thickness.

NOTE 1: Omit eighth for single pair cables only (LSE1SPE-20, LSE1SPE-18, LSE1SPEW-20, and LSE1SPEW-18).

NOTE 2: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Details (non-watertight).

Military part no. M24643/83	Type and size	Conductor size (AWG)	Number of shielded pairs	Shield strand AWG (min.)	Jacket thickness min. avg. (inch)	Cable diameter (inch)	
						min.	max.
-01UO	LSE3SPE-22	22	3	32	0.045	0.415	0.470
-02UO	LSE7SPE-22	22	7	34	0.045	0.500	0.555
-03UO	LSE10SPE-22	22	10	34	0.055	0.635	0.700
-04UO	LSE14SPE-22	22	14	34	0.060	0.695	0.755
-05UO	LSE19SPE-22	22	19	34	0.067	0.770	0.845
-06UO	LSE24SPE-22	22	24	34	0.078	0.920	0.995
-07UO	LSE30SPE-22	22	30	34	0.083	0.975	1.060
-08UO	LSE37SPE-22	22	37	34	0.090	1.060	1.145
-09UO	LSE44SPE-22	22	44	34	0.090	1.200	1.285
-10UO	LSE61SPE-22	22	61	34	0.090	1.340	1.425
-11UO	LSE1SPE-20	20	1	32	0.045	0.300	0.345
-12UO	LSE2SPE-20	20	2	32	0.045	0.430	0.485
-13UO	LSE4SPE-20	20	4	34	0.045	0.475	0.530
-14UO	LSE6SPE-20	20	6	34	0.045	0.555	0.610
-15UO	LSE8SPE-20	20	8	34	0.052	0.605	0.670
-16UO	LSE10SPE-20	20	10	34	0.062	0.715	0.780
-17UO	LSE12SPE-20	20	12	34	0.064	0.735	0.810
-18UO	LSE16SPE-20	20	16	34	0.071	0.825	0.900
-19UO	LSE1SPE-18	18	1	32	0.045	0.310	0.355
-20UO	LSE3SPE-18	18	3	34	0.045	0.455	0.510
-21UO	LSE7SPE-18	18	7	34	0.050	0.580	0.645
-22UO	LSE10SPE-18	18	10	34	0.065	0.750	0.820
-23UO	LSE12SPE-18	18	12	34	0.067	0.775	0.850
-24UO	LSE14SPE-18	18	14	34	0.070	0.820	0.890
-25UO	LSE19SPE-18	18	19	34	0.079	0.930	1.005
-26UO	LSE24SPE-18	18	24	34	0.090	1.095	1.180
-27UO	LSE30SPE-18	18	30	34	0.090	1.165	1.250
-28UO	LSE37SPE-18	18	37	34	0.090	1.265	1.350
-29UO	LSE61SPE-18	18	61	34	0.090	1.600	1.700
-30UO	LSE3SPE-16	16	3	34	0.045	0.515	0.565
-31UO	LSE7SPE-16	16	7	34	0.058	0.680	0.740

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TABLE II. Details (watertight).

Military part no. M24643/83	Type and size	Conductor size (AWG)	Number of shielded pairs	Shield strand AWG (min.)	Jacket thickness min. avg. (inch)	Cable diameter (inch)	
						min.	max.
-32UO	LSE3SPEW-22	22	3	32	0.045	0.450	0.500
-33UO	LSE7SPEW-22	22	7	34	0.045	0.550	0.600
-34UO	LSE10SPEW-22	22	10	34	0.060	0.700	0.760
-35UO	LSE14SPEW-22	22	14	34	0.073	0.760	0.830
-36UO	LSE19SPEW-22	22	19	34	0.086	0.860	0.930
-37UO	LSE24SPEW-22	22	24	34	0.090	1.005	1.090
-38UO	LSE30SPEW-22	22	30	34	0.090	1.065	1.150
-39UO	LSE37SPEW-22	22	37	34	0.090	1.160	1.245
-40UO	LSE44SPEW-22	22	44	34	0.090	1.310	1.395
-41UO	LSE61SPEW-22	22	61	34	0.090	1.455	1.560
-42UO	LSE1SPEW-20	20	1	32	0.045	0.325	0.370
-43UO	LSE2SPEW-20	20	2	32	0.045	0.465	0.520
-44UO	LSE4SPEW-20	20	4	34	0.045	0.515	0.565
-45UO	LSE6SPEW-20	20	6	34	0.053	0.610	0.675
-46UO	LSE8SPEW-20	20	8	34	0.057	0.665	0.730
-47UO	LSE10SPEW-20	20	10	34	0.067	0.780	0.855
-48UO	LSE12SPEW-20	20	12	34	0.069	0.805	0.880
-49UO	LSE16SPEW-20	20	16	34	0.077	0.910	0.985
-50UO	LSE1SPEW-18	18	1	32	0.045	0.345	0.390
-51UO	LSE3SPEW-18	18	3	34	0.045	0.520	0.570
-52UO	LSE7SPEW-18	18	7	34	0.059	0.680	0.745
-53UO	LSE10SPEW-18	18	10	34	0.075	0.885	0.960
-54UO	LSE12SPEW-18	18	12	34	0.078	0.915	0.990
-55UO	LSE14SPEW-18	18	14	34	0.082	0.960	1.045
-56UO	LSE19SPEW-18	18	19	34	0.090	1.080	1.165
-57UO	LSE24SPEW-18	18	24	34	0.090	1.280	1.365
-58UO	LSE30SPEW-18	18	30	34	0.090	1.360	1.445
-59UO	LSE37SPEW-18	18	37	34	0.090	1.470	1.570
-60UO	LSE61SPEW-18	18	61	34	0.090	1.865	1.970
-61UO	LSE3SPEW-16	16	3	34	0.045	0.550	0.605
-62UO	LSE7SPEW-16	16	7	34	0.063	0.735	0.810

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	
22 AWG	17.37
20 AWG	11.03
18 AWG	7.47
16 AWG	5.10
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,500
Conductor to shield	1,500
Component shield to all other shields	500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	2,500
Conductor to shield	2,500
Component shield to all other shields	100
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Capacitance unbalance (percent, max.)	8
Characteristic impedance, at 1 megahertz (ohms)	75±5

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50

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Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (applicable to watertight constructions only) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	75
Elongation (percent, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, construction, and coverage)	No failure
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability (applicable to watertight constructions only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	5.0
Electromagnetic pulse (EMP) response (dB, min.)	80

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Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (86±3.6 °F [30±2 °C])	No damage
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure
Long-term jacket immersion	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All	500

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-004)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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